**In-Depth Research Report on Aerodynamic and Structural Parameters of the ATR 42-600 Aircraft for Mathematical Modeling and V-n Diagram Analysis**

Designing, analyzing, and evaluating the structural performance of commercial aircraft according to international standards is a highly precise engineering process. This is especially true for compliance with the Airworthiness Requirements under the CS-25 standard of the European Union Aviation Safety Agency (EASA). This report compiles and deeply analyzes the parameters of the ATR 42-600, a regional turboprop passenger aircraft featuring the New Avionic Suite (NAS) or Mod 5948.

The primary objective of analyzing these parameters is to prepare the Input Data Structure for Stage 1 (Data Collection & Parameter Definition) to be used in MATLAB. The focus is on generating the V-n Diagram (Velocity-load factor diagram) and solving equations for the Balancing Tail Load, which represents a critical structural condition that the Horizontal Tail Plane (HTP) must endure. This analysis not only presents statistical figures but also delves into the aerodynamic interactions and the consequent impacts each variable has on the overall structural safety of the aircraft.

**1. Weight & Mass Limits Analysis and Structural Impacts**

In aeronautical engineering, the mass and weight of an aircraft are the primary factors determining the magnitude of inertia, aerodynamic forces, and the stresses the main structure must handle. Each weight limit specified in the EASA Type Certificate Data Sheet (TCDS) No. A.084 plays a different role in simulating boundary conditions in MATLAB.

The Maximum Take-Off Weight (MTOW) of the ATR 42-600 is standardized at 18,600 kg. This MTOW is the primary mass parameter for determining the peaks of the V-n Diagram. When the aircraft enters a maneuvering state at a maximum load factor (e.g., +2.5g according to commercial standards), the main wing must generate lift equal to 2.5 times the MTOW. Calculating loads at MTOW determines the ultimate strength limit of the Wing-to-Fuselage Attachment and serves as the reference mass for climb performance and maximum engine power usage.

However, regarding stress analysis and the Wing Root Bending Moment, the most critical condition does not occur at MTOW but rather at the Maximum Zero Fuel Weight (MZFW), which is limited to 17,000 kg. This phenomenon is explained by the principle of Inertial Relief. Normally, when the wings are full of fuel, the fuel's weight creates a downward inertial force that counteracts the upward aerodynamic lift, reducing the bending moment at the wing root. But when the aircraft operates at MZFW (carrying maximum passenger and cargo payload but with empty wing tanks), the wing loses this resisting force, causing all lift to act directly on the wing root. Writing a MATLAB script to verify structural integrity must clearly separate calculation loops for MTOW and MZFW to find the most severe bending moment.

The Operational Empty Weight (OEW) for this model ranges from approximately 11,500 to 11,700 kg, with a typical value of 11,600 kg. This includes the basic structure, propulsion systems, and undrainable fluids. OEW serves as the base vector for calculating mass distribution when evaluating the Center of Gravity (C.G.). Adding payload and fuel into the MATLAB model requires calculating moments around a reference point to locate the C.G. in various flight conditions.

Additionally, the Maximum Landing Weight (MLW) of 18,300 kg is vital for analyzing dynamic loads when the nose and main landing gears impact the runway. This impact creates a sudden pitch-down moment, representing another condition the horizontal tail plane must withstand.

| Weight Limits | Value (kg) | MATLAB Modeling Role |
|---|---|---|
| Maximum Take-Off Weight (MTOW) | 18,600 | Used to evaluate Max Lift and target area of V-n Diagram in fully loaded condition |
| Maximum Landing Weight (MLW) | 18,300 | Analyzes landing gear impact forces and dynamic pitch-down moment |
| Maximum Zero Fuel Weight (MZFW) | 17,000 | Used to find the critical condition for wing root bending moment due to the lack of Inertial Relief |
| Operational Empty Weight (OEW) | ~11,600 | Serves as the Base Mass for weight distribution to determine the center of gravity |

**2. Main Wing Geometry and Aerodynamics**

The shape and proportions of the main wing act as a medium translating speed and angle of attack into aerodynamic forces. The Wing Reference Area ($S$ or $S_w$) of the ATR 42-600 is 54.5 square meters. With a Wing Span ($b$) of 24.57 meters, the Aspect Ratio ($AR = b^2/S$) is calculated to be approximately 11.08. This relatively high aspect ratio is a defining characteristic of regional aircraft focused on fuel efficiency, as it significantly reduces Induced Drag during cruise flight.

A characteristic profoundly affecting the Balancing Tail Load calculation is the airfoil selection. This aircraft uses a blended airfoil section: the RA 1843 (NACA 43018 mod) at the wing root, and the RA 1343 (NACA 43013 mod) at the wing tip. The code 43018 reflects a modification of the NACA 5-digit series with a maximum thickness of 18% of the chord at the root. This thickness provides adequate volume for fuel tanks and increases the polar moment of inertia for the wing spar to resist bending forces.

However, the aerodynamic consequence of using a highly cambered airfoil is the generation of a noticeably negative Zero-lift Pitching Moment Coefficient ($C_{m0}$). The wing's natural nose-down tendency forces the horizontal tail plane to produce a downward tail load in the opposite direction to maintain equilibrium. Estimating $C_{m0}$ from airfoil characteristics in computer software acts as a constant offset in the moment equilibrium equation around the center of gravity.

**3. C.G. Envelope & Tail Moment Arm**

Longitudinal Static Stability and handling qualities are bounded by the movement of the Center of Gravity relative to the Mean Aerodynamic Chord (MAC). For the ATR 42 series, the MAC ($\bar{c}$) length is specified as 2.285 meters (89.96 inches). The approved C.G. range is from 16.8% to 32.5% of the MAC. This wide envelope accommodates variations in passenger and cargo placement during commercial flights.

In terms of structural design and MATLAB analysis, the Forward C.G. Limit at 16.8% MAC is considered the worst-case scenario or Critical Load Case for stresses on the horizontal tail. The engineering mechanics can be explained as follows: generally, total wing lift is referenced at the aerodynamic center, approximately 25% of the MAC. When the C.G. shifts forward to 16.8% MAC, the aircraft's weight pressing down at the C.G. creates a pitch-down moment around the 25% MAC point. Combined with the wing's negative $C_{m0}$, the Horizontal Tail Plane (HTP) must generate an amplified downward force to maintain level flight. The higher the aircraft speed, the more the dynamic pressure magnifies the wing's pitching moment, causing the Balancing Tail Load to reach its maximum negative value at the Forward C.G. Limit. Inputting this parameter in MATLAB's calculation loops is indispensable for finding the HTP's critical stress condition.

Conversely, the Aft C.G. Limit at 32.5% MAC relates to stability requirements and the Static Margin. As the C.G. moves backward towards the entire aircraft's Neutral Point, the aircraft becomes more responsive to pitch controls but loses its inherent restoring moment. The 32.5% limit prevents the C.G. from moving behind the Neutral Point, which would lead to longitudinal instability. This data is used to establish the bottom-right performance boundary of the MATLAB equations.

The longitudinal distance from the center of gravity to the tail's aerodynamic center (Tail Moment Arm, $l_t$) serves as a crucial divisor in the equilibrium equation. The $l_t$ of the ATR 42-600 is approximately 11.46 meters. A moment arm of this length enables the HTP to generate restoring moments without requiring an excessively large tail area (HTP area is approximately 157.7 sq ft or 14.65 sq m). Reducing HTP area lowers tail structure weight and Wetted Area, directly correlating to reduced Skin Friction Drag during long-haul flights.

**3.1 Horizontal Tail Plane (HTP) Geometry, Aerodynamics, and Parameter Creation**

To effectively model the horizontal tail plane for V-n diagram and balancing tail load calculations, precise geometric and aerodynamic parameters must be established. These inputs are often structured utilizing specific parameter blocks like HTPLNF (Horizontal Tail Planform) in computational engineering tools.

**Airfoil Selection and Aerodynamic Behavior:** Unlike the highly cambered main wing, the ATR 42 utilizes a NACA 0012 airfoil for both the root and the tip of the horizontal tail plane. The NACA 0012 is a completely symmetrical airfoil profile. This symmetry is a crucial aerodynamic design choice because its zero-lift pitching moment coefficient ($C_{m0}$) is exactly zero, ensuring the tail does not generate its own intrinsic rotational force. For a symmetrical wing like the NACA 0012, lift generation is strictly proportional to its effective angle of attack, producing exactly zero lift at a zero geometric angle of attack. This predictable, linear aerodynamic behavior allows the HTP to efficiently generate either the upward or downward balancing loads required to counteract the varying pitching moments of the main wing.

**Geometric Dimensions (HTPLNF):** When defining the HTP planform parameters for MATLAB analysis, specific physical bounds must be utilized. The ATR 42 HTP has a span of approximately 7.25 meters. Its aspect ratio falls between 4.50 and 4.77, while the reference planform area varies slightly based on the specific structural baseline model, typically ranging from 11.03 square meters up to 14.65 square meters (126.3 to 157.7 sq ft).

| Parameter / Characteristic | Value / Description | Note / MATLAB Application |
|---|---|---|
| Airfoil Section | NACA 0012 (Symmetrical) | Ensures linear lift without intrinsic pitching moment. |
| Zero-lift Pitching Moment ($C_{m0}$) | 0 | Constant used in the equilibrium equation for the tail. |
| Planform Area | 11.03 - 14.65 sq.m (126.3 - 157.7 sq.ft) | Defined in HTPLNF block; scales the balancing load capacity. |
| Span | 7.25 m | Defined in HTPLNF block. |
| Aspect Ratio | 4.50 - 4.77 | Affects induced drag and lift curve slope of the tail. |

**Longitudinal Stability and C.G. Envelope Integration:** The HTP is the primary aerodynamic surface ensuring longitudinal static stability. Mathematically, the required condition for this stability is that the first derivative of the aircraft's pitching moment coefficient with respect to the angle of attack must be negative ($\frac{\partial C_M}{\partial \alpha} < 0$). This constraint ensures that any disturbance increasing the angle of attack produces a restoring nose-down pitching moment.

The overall effectiveness of the HTP is intrinsically linked to the Stability Margin (SM), defined as the normalized difference between the aircraft's center of gravity (C.G.) and the Neutral Point (NP): $SM = x_{CG} - x_{NP}$. For a longitudinally stable aircraft, the C.G. must remain forward of the NP, resulting in a negative SM. When operating at the extreme Forward C.G. Limit (16.8% MAC), the downward leverage of the aircraft's weight is at its absolute maximum. Consequently, the HTP's NACA 0012 airfoil must operate at a highly negative effective angle of attack (which is further influenced by the main wing's downwash and the propeller's slipstream) to generate the ultimate downward balancing load required to maintain equilibrium.

**4. Design Speeds for V-n Diagram according to CS-25**

Simulating the flight envelope with a V-n Diagram illustrates the relationship between aircraft velocity (X-axis) and load factor (Y-axis). The EASA CS 25.333 standard mandates that manufacturers meticulously define these boundaries to prevent structural failure. Inputting data into MATLAB requires establishing the following velocity vectors:

1. **Stall Speed ($V_{S1}$)** The stall speed in a clean configuration (flaps and gear up) marks the leftmost boundary of the V-n Diagram curve. The maximum lift coefficient ($C_{L,max}$) for the ATR 42's clean wing is generally around 1.5. Calculated at MTOW and sea-level air density, $V_{S1}$ is approximately

110-120 KIAS. The curve area on the left side of the V-n Diagram represents an aerodynamic limitation, meaning the aircraft cannot generate higher load factors above this curve because the wing will stall before the structure takes heavy loads.

2. **Design Maneuvering Speed ($V_A$)** $V_A$ is a paramount structural safety parameter for pilots. It can be theoretically calculated from the equation $V_A = V_{S1}\sqrt{n_z}$, where $n_z$ is the critical positive load factor (+2.5g for commercial aircraft). The engineering significance of $V_A$ is that it is the maximum speed at which the pilot can safely apply full, abrupt control deflection. If an emergency occurs below $V_A$, the aircraft will stall before the load factor destroys the structure (stall before break). In MATLAB, the intersection of the maximum lift curve and the $n_z = 2.5$ line represents the coordinate for $V_A$.

3. **Design Cruising Speed ($V_C$)** $V_C$ represents the primary operational point at cruise altitude. For the ATR 42-600, the Maximum Operating Limit Speed ($V_{MO}$) is certified by EASA at 250 KIAS (historically or in some modifications at 240/250 KIAS). Often in analysis, $V_C$ is set equal to or slightly below $V_{MO}$. The CS-25 standard requires a Gust Loads analysis at $V_C$ using a 50 ft/sec gust criteria, which is crucial for designing wing spars to withstand turbulent weather.

4. **Design Dive Speed ($V_D$ or $V_{DF}$)** To provide a safety margin against loss of control or sudden dives, $V_D$ is always set higher than $V_C$. Under CS 25.335(b), $V_D$ usually must have at least a 1.25 margin over $V_C$ in some aircraft families, or is otherwise limited by Compressibility Effects (Mach Number limits). The MATLAB V-n Diagram simulation uses $V_D$ to close the rightmost edge of the envelope. At this point, the horizontal tail plane experiences the maximum Balancing Tail Load because dynamic pressure is at its peak, amplifying the pitch-down moment from $C_{m0}$ to its most severe extent.

| **CS-25 Standard Speeds** | **Role and Calculation in V-n Diagram** |
|---|---|
| Stall Speed ($V_{S1}$) | Initiates the aerodynamic lift limit curve on the left side |
| Maneuvering Speed ($V_A$) | Defines the top-left intersection at +2.5g, preventing over-maneuvering structural damage |
| Cruising / Max Op Speed ($V_C/V_{MO}$) | 250 KIAS - Primary operational point and reference for 50 ft/sec gust load analysis |
| Dive Speed ($V_D$) | Closes the right side of the diagram, marking where the tail sustains maximum negative balancing load |

**5. MATLAB Algorithms and Equations for Balancing Tail Load Calculation**

The ultimate goal of gathering these parameters is to create an accurate computer script for structural evaluation. The ATR 42-600's horizontal tail must not only provide stability but also support the Balancing Tail Load (BTL or $L_H$) to offset the aircraft's unbalanced moments.

In a steady-state maneuver, pitching acceleration is zero. The reaction forces on the Z-axis must balance the weight multiplied by the load factor ($n_z W$). Finding $L_H$ starts with summing the moments around the center of gravity ($\Sigma M_{cg} = 0$). Assuming horizontal drag and engine thrust moment arms are negligible, the equation takes the form:

$$L_H = \frac{\left[\left(X_{cg} - 0.25\right)C_{L\alpha} + C_M\right]qS_w c_w}{l_h} + \frac{M_H}{l_h}$$

For MATLAB programming, the BTL equation is usually expressed using a Linearized Aerodynamic Matrix:

$$BTL = \frac{n_z W\left(X_{CG} - 0.25 + \frac{dC_M}{dC_L}\right) + C_{m0}qS_w\bar{c}}{l_h}$$

The algorithmic process in MATLAB consists of these main components:

1. **State-space Matrix Initialization:**

The program generates a weight vector $W$ from OEW (11,600 kg) to MTOW (18,600 kg), a $X_{CG}$ vector from 0.168 to 0.325, and sweeps through a velocity range $V$ from $V_S$ to $V_D$ to calculate dynamic pressure $q = 0.5\rho V^2$ across different altitudes.

2. **Static Critical Point Detection:** The equation clearly shows that the term $(X_{CG} - 0.25)$ reaches its maximum negative value at the extreme forward center of gravity $X_{CG} = 0.168$. This difference of -0.082, multiplied by the critical load factor $n_z = 2.5$ at MTOW, creates a massive nose-down moment. Additionally, $C_{m0}$ from the cambered airfoil accumulates further negative moment. As the aircraft operates at high speeds like $V_D$ (creating massive $q$), the $BTL$ plotted by the program will represent the ultimate negative lift. The HTP structure, hinge mounts, and trim tabs must be evaluated for fatigue based on this peak value.

3. **Dynamic Maneuvering Injection:** Under CS 25.331, the aircraft must withstand an abrupt pitching maneuver. In this scenario, angular acceleration is non-zero, and the pilot abruptly pulls the elevator. The Elevator Deflection Limit ($\Delta\delta_{emax}$) for this class of aircraft is typically designed not to exceed 25 degrees to prevent flow separation or tail stall.

The dynamic load case is an adjustment adding aerodynamic forces from the deflected surfaces plus angular acceleration:

$$\Delta L_{H\theta} = L_{\alpha_h}\Delta\alpha_h + L_{\delta e}\Delta\delta_{emax}$$

The MATLAB plots in this mode will display a fractional-second peak load that is typically higher than the steady-state load, often representing the most critical design condition for sizing the tail spars.

4. **Downwash and Slipstream Modifiers:** An additional challenge for turboprop aircraft is that the propeller slipstream alters the dynamic pressure ratio at the tail ($\eta = q_H/q$) compared to glide

flight. The PW127M engines add energy to the airflow, increasing $q_H$. Simultaneously, the massive lift from the main wings generates downwash ($\epsilon$), altering the relative angle of attack at the tail. The MATLAB program must incorporate these correction terms into the coefficients ($C_{L,H}$) to ensure the model closely mimics wind tunnel testing behaviors.

**6. Analytical Syntheses**

Decoding the in-depth parameters of the ATR 42-600 to serve as foundational data for EASA CS-25 structural analysis reveals a unified engineering art that interconnects all subsystems. Weight limits at 18,600 kg (MTOW) and 17,000 kg (MZFW) establish clear boundaries for wing root bending forces, while the NACA 43018 mod and NACA 43013 mod airfoils offer exceptional short takeoff performance at the cost of generating an inherently negative $C_{m0}$ pitching moment.

When combining the geometric dimensions of the wing area (54.5 sq.m.), reference chord (2.285 m), and horizontal tail moment arm (11.46 m) with the forward C.G. limit of 16.8% MAC inside MATLAB, the V-n Diagram can scale the envelope boundary from stall speed ($V_S$) to the maximum dive speed ($V_D$). Structuring this data through equilibrium equations explicitly highlights the immense downward forces the tail structure must endure during dives and abrupt maneuvers. The MATLAB output generated in this Stage 1 process ultimately forms the supporting foundation for safety certification, fatigue limit evaluation, and ultimate structural failure analysis of the aircraft.